

PROGRAMMING FUNDAMENTALS

Spring 2024

10th Mar

Converting decimal Numbers to Integers

- Assigning a floating-point value to an integer variable → the fractional part is discarded

```
double price = 2.55;
```

```
int dollars = price; // Sets dollars to 2
```

- Often, you want to round to the *nearest* integer
- Add 0.5 and then convert to an integer

```
int dollars;
```

```
dollars = price + 0.5; // Rounds to the nearest integer
```

- 2.55 is turned into 3.05, which is then truncated to 3.
(For a negative floating-point value, you subtract 0.5)

Infinite Loop

- loops must contain within themselves a way to terminate
- If a loop does not have a way of stopping, it is called an infinite loop
- An *infinite loop* continues to repeat until the program is interrupted

```
int number = 0;  
while (number < 5)  
{  
    cout << "Hello\n";  
}
```

Infinite Loop

```
int number = 0;  
while (number < 5); // This semicolon is an ERROR!  
{  
    cout << "Hello\n";  
    number++;  
}
```

- The semicolon at the end of the first line is assumed to be a null
- To the compiler, this loop looks like:

```
while (number < 5);
```

- This while loop will forever execute the null statement, which does nothing

Infinite Loop

```
int number = 0;  
// This loop is missing its braces!  
while (number < 5)  
cout << "Hello\n";  
number++;
```

- `number++` statement is not in the body of the loop
- No braces → the while statement only executes the statement that immediately follows it
- This loop will execute infinitely because there is no code in its body that changes the `number` (loop variable)

Ex-1 (show output)

```
int n = 5;  
while (n >= 0)  
{  
    n--;  
    cout << n << endl;  
}
```



D:\Books-ICT\F2020\L

4

3

2

1

0

-1

Process exited after

Press any key to co

Ex-2

```
int n = 1;  
while (n <= 3)  
{  
    cout << n << ", ";  
    n++;  
}
```




D:\Books-ICT\F2020\Lec-

1, 2, 3,

Process exited after
Press any key to con

Ex-3

- a is 2 and n is 4

```
int r = 1;  
int i = 1;  
while (i <= n)  
{  
    r = r * a;  
    i++;  
}
```

Ex-3

- a is 2 and n is 4

```
int r = 1;  
int i = 1;  
while (i <= n)  
{  
    r = r * a;  
    i++;  
}
```

NO output

Ex-4

- Write programs with loops that compute
 - a. The sum of all even numbers between 2 and 100 (inclusive).
 - b. The sum of all squares between 1 and 100 (inclusive).
 - c. All powers of 2 from 20 up to 220.
 - d. The sum of all odd numbers between a and b (inclusive), where a and b are inputs.
 - e. The sum of all odd digits between a given range. User is asked to enter the start and the end points.

for Loop

```
for (counter = 1; counter <= 10; counter++)  
{  
    cout << counter << endl;  
}
```

- Another commonly-used term for a counter-controlled loop is *definite* loop

```
for (counter = 1; counter <= 10; counter++)  
{  
    cout << counter << endl;  
}
```

- This loop groups the initialization, condition, and update expressions together
1. Initialize counter
 2. Check condition
 3. Run Loop body
 4. Update counter
 5. Check condition again

Ex-5

```
int main ()  
{  
    int sum = 0;  
    int counter;  
    for (counter = 0; counter <= 10; counter = counter + 2)  
    {  
        sum = sum + counter;  
    }  
    cout<<"The final value in sum = "<<sum;  
    return 0;  
}
```

Ex-5 (output)

```
The final value in sum = 30
-----
Process exited after 0.1286 seconds with return value 0
Press any key to continue . . .
```


Ex-6

```
int main ()
{
    int sum = 0;
    int counter;
    cout<<"Values in counter variable"<<endl;
    for (counter = 0; counter <= 10; counter = counter + 2)
    {
        cout<<counter<<endl;
        sum = sum + counter;
    }
    cout<<"The final value in sum = "<<sum;
    return 0;
}
```

Ex-6 (output)

```
Values in counter variable
```

```
0
```

```
2
```

```
4
```

```
6
```

```
8
```

```
10
```

```
The final value in sum = 30
```

```
-----
```

```
Process exited after 0.1387 seconds with return value 0
```

```
Press any key to continue . . .
```

Ex-3

- The initial balance in your bank account is Rs. 10,000
- The interest rate per year is 10%
- You need to ask the user the number of years he/she wants to invest in the bank
- Your program should display the initial balance and yearly balance for the entered number of years using a loop

Ex-3

Your Starting Balance = 10000

Enter the number of years for investing your money = 3

Current Balance = 11000

Current Balance = 12100

Current Balance = 13310

Ex- 4

- Change the program to display the output on the next page

Ex-4 (output)



D:\Books-ICT\F2020\Lec22-Ex3.exe

Enter number of years: 5

1	11000
2	12100
3	13310
4	14641
5	16105.1

Process exited after 1.186 seconds
Press any key to continue . . .

Ex-4 (using while loop for Ex-3)

```
int main()
{
    const double start_bal = 10000;
    double balance = start_bal;
    int years;
    cout << "Enter number of years: ";
    cin >> years;
    int i;
    i = 1;
    while(i <= years)
    {
        balance =balance * (1 + 0.1);
        cout << i << "          "<<balance << endl;
        i++;
    }
    return 0;
}
```

Ex-5

```
#include <iostream>
using namespace std;

int main ()
{
    int even;
    cout<<"List of Even Numbers"<<endl;
    for (even = 2; even <= 20; even = even + 2)
    {
        cout<<even<<endl;
    }
    return 0;
}
```


Ex-5 (output)

```
List of Even Numbers
2
4
6
8
10
12
14
16
18
20

-----
Process exited after 0.1332 seconds
Press any key to continue . . .
```

Ex-6

- Different courses have different number of quizzes.
- Write a program that will keep on accepting the marks in a quiz (out of 10) until the user enters -1
- sum is calculated and displayed
- Average is calculated and displayed

Ex-6 (error?????)

```
int main()
{
    float marks;
    float sum;
    float average;
    sum = 0;
    average = 0;

    cout << "Enter marks in Quiz 1:  (-1 to exit)";
    cin >> marks;

    while(marks != -1)
    {
        sum = sum + marks;
        cout << "Enter marks in next Quiz :  (-1 to exit)";
        cin >> marks;
    }
    cout<<"Total Score in quizzes =  "<<sum<<endl;
    cout<<"Average score in quizzess =  "<<average;
    return 0;
}
```

Ex-6 (output)



D:\Books-ICT\F2020\Lec22-Ex6.exe

```
Enter marks in Quiz 1:  (-1 to exit)10
Enter marks in next Quiz :  (-1 to exit)5
Enter marks in next Quiz :  (-1 to exit)5
Enter marks in next Quiz :  (-1 to exit)10
Enter marks in next Quiz :  (-1 to exit)-1
Total Score in quizzes = 30
Average score in quizzess = 0
-----
Process exited after 12.8 seconds with return
Press any key to continue . . .
```

Ex-6 (corrected version)

```
int main()
{
    float marks;
    float sum;
    float average;
    sum = 0;
    average = 0;
    int counter;
    counter = 0;
    cout << "Enter marks in Quiz 1:  (-1 to exit)  ";
    cin >> marks;
    while(marks != -1)
    {
        counter = counter + 1;
        sum = sum + marks;
        cout << "Enter marks in next Quiz :  (-1 to exit)  ";
        cin >> marks;
    }
    average = sum/counter;
    cout<<"Total Score in quizzes =  "<<sum<<endl;
    cout<<"Average score in quizzess =  "<<average;
    return 0;
}
```

Ex-6 (revised output)

 D:\Books-ICT\F2020\Lec22-Ex6.exe

```
Enter marks in Quiz 1: (-1 to exit) 10
Enter marks in next Quiz : (-1 to exit) 5
Enter marks in next Quiz : (-1 to exit) 5
Enter marks in next Quiz : (-1 to exit) 10
Enter marks in next Quiz : (-1 to exit) -1
Total Score in quizzes = 30
Average score in quizzess = 7.5
-----
Process exited after 8.826 seconds with return value 0
Press any key to continue . . .
```

Pre-Test Loops

- `while` and `for` loops are *pre-test* loops
- The condition is tested before entering the loop body

Post-Test Loop example

- to execute the body of a loop at least once and perform the loop test after the body is executed

```
int value;  
do  
{  
    cout << "Enter a value < 100: ";  
    cin >> value;  
}  
while (value >= 100);
```


Ex-7 : code to display this

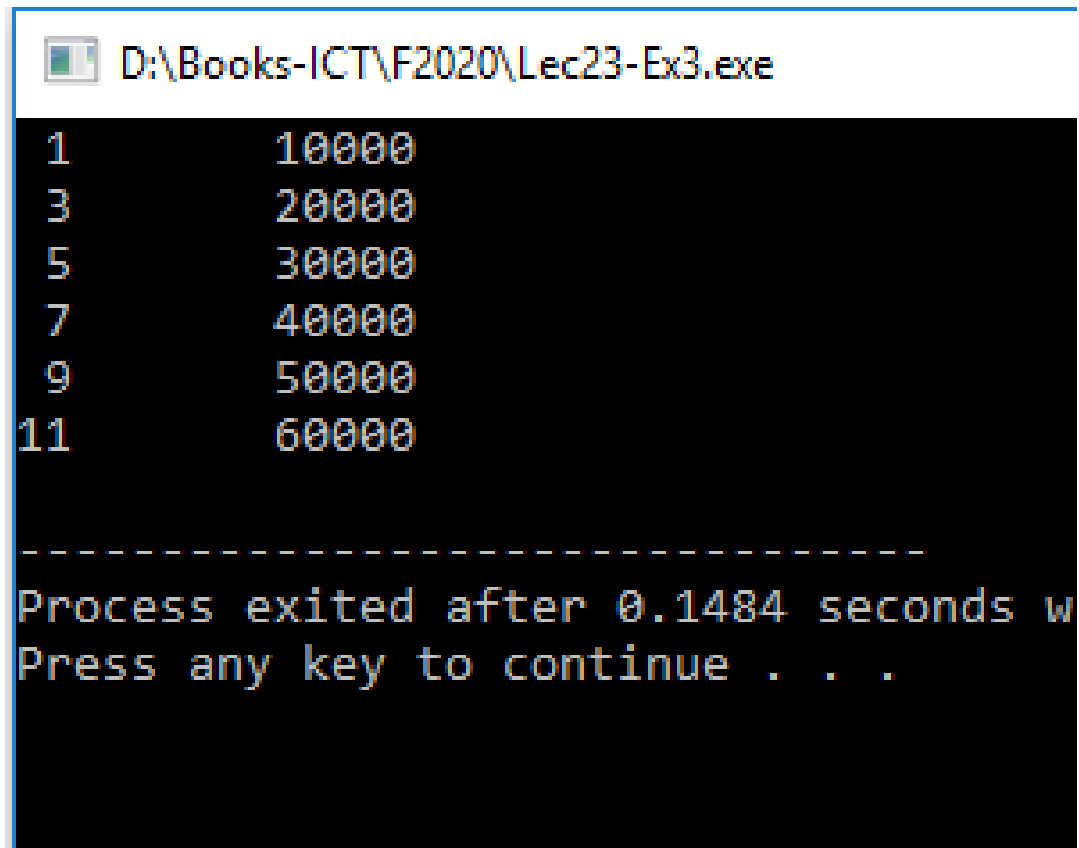


The screenshot shows a Windows command prompt window with the title bar "D:\Books-ICT\F2020\Lec23-Ex2.exe". The output of the program is as follows:

```
//////////  
+-----+  
( |   o   o   | )  
|               |  
|               |  
|               |  
|               |  
+-----+  
  
-----  
Process exited after 0.121 se  
Press any key to continue . .
```

Ex-8

- Write a **for** loop to print the following:



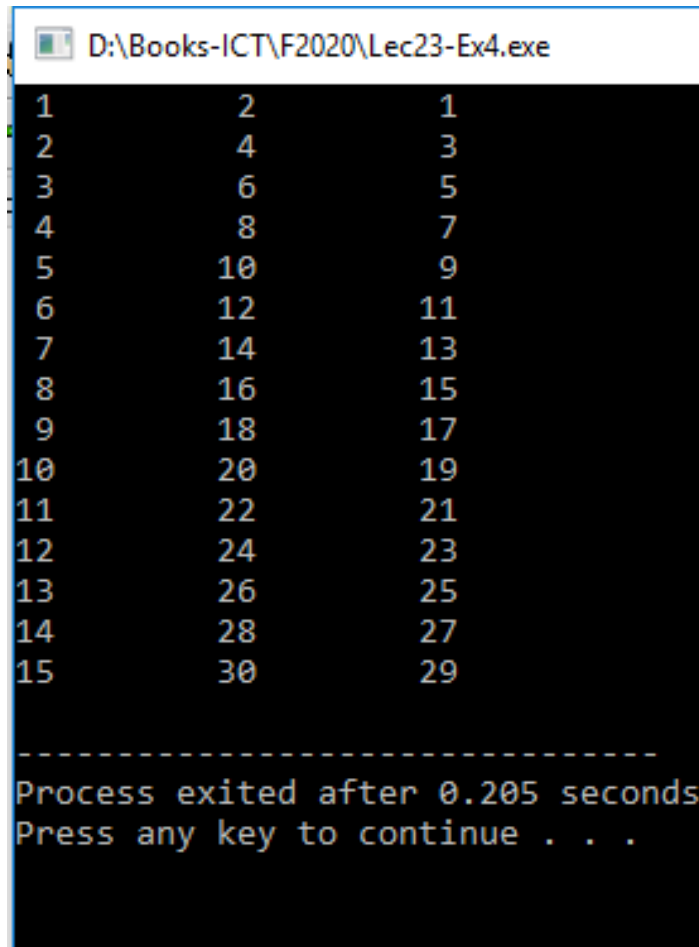
```
D:\Books-ICT\F2020\Lec23-Ex3.exe

1      10000
3      20000
5      30000
7      40000
9      50000
11     60000

-----
Process exited after 0.1484 seconds w
Press any key to continue . . .
```

Ex-9

- Use one **for** loop to display the following:



```
D:\Books-ICT\F2020\Lec23-Ex4.exe
1      2      1
2      4      3
3      6      5
4      8      7
5     10      9
6     12     11
7     14     13
8     16     15
9     18     17
10    20     19
11    22     21
12    24     23
13    26     25
14    28     27
15    30     29

-----
Process exited after 0.205 seconds
Press any key to continue . . .
```

Ex-10

```
#include <iostream>
#include<iomanip>
#include<cstdlib>
using namespace std;

int main ()
{
    int i;
    int r_num;

    for(i= 1;i<=10;i++)
    {
        r_num = rand();
        cout<<r_num<<endl;
    }
    return 0;
}
```

Try running the code and
you will find it is
generating same numbers
every time

The rand function is declared
in the <cstdlib> header

Solution

- Specify a *seed* for the random number sequence
- The seed is set with the `srand` function
- A simple value to use as a seed is the current time:
`srand(time(0));`
- include the `<ctime>` header that declares the time function

Ex-11

```
#include <iostream>
#include<iomanip>
#include<cstdlib>
#include<ctime>
using namespace std;

int main ()
{
    srand(time(0));
    int i;
    int r_num;

    for(i= 1;i<=10;i++)
    {
        // to generate ranom numbers from 1 to 6
        r_num = rand() % 6 + 1;
        cout<<r_num<<endl;
    }
    return 0;
}
```

 D:\Books-ICT\F2020\Lec23-Ex6.exe

4
3
2
5
6
3
3
4
6
6

Process exited after 0.2154 s
Press any key to continue . .

Book Reference

- D.S. Malik (ch - 5)
- Tony Gaddis (ch - 5)